XIAMEN UNIVERSITY MALAYSIA

ASSIGNMENT 1 - Marking Scheme

1. (2 points)

(a) No. This is simple accounting.

(b) Yes. We would attempt to create a model that can predict the continuous value of the stock price. This is an example of the area of data mining known as predictive modelling.

2. (3 points)

(a) binary, ordinal

(b) discrete, ordinal

(c) continuous, ratio

3. (2 points)

(a)

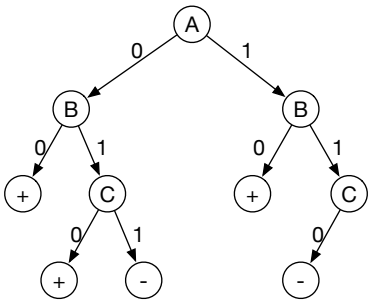
$D(\mathbf{x},\mathbf{y})=\sqrt{(1-2)\times(1-2)+(2-4)\times(2-4)+(1-2)\times(1-2)+(2-4)\times(2-4)}=\sqrt{10}$;

(b)

$D(\mathbf{x},\mathbf{y})=\sqrt{(0-2)\times(0-2)+(1-0)\times(1-0)+(0-1)\times(0-1)+(2-0)\times(2-0)}=\sqrt{10}$;

4. (3 points)

Constructed decision tree:



Predicted class labels:

Validation:

Instance	A	B	C	Class
11	0	0	0	+
12	0	1	1	–
13	1	1	0	–
14	1	0	1	+
15	1	0	0	+